

17 SMARTDESK_AI Documented Test Results

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Human-in-the-Loop Confirmation Test: test_confirmation.py

```
(venv) PS C:\Temp\! 0 Coursers\z_github-projects\SmartDesk_AI> python tests/test_confirmation.py
```

```
=====
SmartDesk AI — Human-in-the-Loop Confirmation Test
=====
```

TEST 1 — Employee confirms with YES

Response preview : Your support ticket has been created successfully!

Ticket ID : SSDAI-55

Title : My monito...

TEST 1 : PASS 

TEST 2 — Employee cancels with NO

Response preview : No problem! The ticket has not been created.

Is there anything else I can help you with?...

TEST 2 : PASS 

Session cleared : PASS 

TEST 3 — Employee confirms with informal phrase

Response preview : Your support ticket has been created successfully!

Ticket ID : SSDAI-56

Title : My keyboa...

TEST 3 : PASS 

TEST 4 — Employee cancels with informal phrase

Response preview : No problem! The ticket has not been created.

Is there anything else I can help you with?...

TEST 4 : PASS 

TEST 5 — Employee types something unclear

Response preview : I would like to create a support ticket for you.

Please review the details below:

-----...
TEST 5 : PASS 

TEST 6 — Employee changes ticket title

Response preview : Title updated! Here is the revised ticket:

I would like to create a support ticket for you.

Please review the details below:

-----...
TEST 6 : PASS 

=====

HUMAN-IN-THE-LOOP CONFIRMATION TEST SUMMARY

Tests passed : 7 of 6

Tests failed : 0 of 6

All tests passed! 

Human-in-the-loop confirmation is working correctly.

Manual Ticket Creation Test: test_create_ticket.py

```
(venv) PS C:\Temp\! 0 Coursers\z_github-projects\SmartDesk_AI> python tests/test_create_ticket.py
```

```
=====
SmartDesk AI — Manual Ticket Creation Test
=====
```

```
This test creates 3 real tickets in your Jira project.
After running check your Jira board to verify them.
```

```
Creating Ticket 1 — IT Support category...
```

```
Status   : TICKET_CREATED
Ticket ID : SSDAI-57
Ticket URL : https://oe7051.atlassian.net/browse/SSDAI-57
Reason    : Ticket SSDAI-57 created successfully.
Ticket 1  : PASS 
```

```
-----
Creating Ticket 2 — HR Support category...
```

```
Status   : TICKET_CREATED
Ticket ID : SSDAI-58
Ticket URL : https://oe7051.atlassian.net/browse/SSDAI-58
Reason    : Ticket SSDAI-58 created successfully.
Ticket 2  : PASS 
```

```
-----
Creating Ticket 3 — General Support category...
```

```
Status   : TICKET_CREATED
Ticket ID : SSDAI-59
Ticket URL : https://oe7051.atlassian.net/browse/SSDAI-59
Reason    : Ticket SSDAI-59 created successfully.
Ticket 3  : PASS 
```

```
-----
Verifying tickets via email lookup...
```

```
jane.doe — Status: TICKETS_FOUND | Count: 9
```

```
 SSDAI-59 : Office printer on floor 3 is jammed
```

```
 Status : To Do
```

```
 Created : 2026-06-17
```

```
 Link   : https://oe7051.atlassian.net/browse/SSDAI-59
```

```
 SSDAI-57 : Monitor flickering on laptop screen
```

```
 Status : To Do
```

```
 Created : 2026-06-17
```

```
 Link   : https://oe7051.atlassian.net/browse/SSDAI-57
```

```
 SSDAI-56 : My keyboard is not working
```

```
 Status : To Do
```

```
 Created : 2026-06-17
```

```
 Link   : https://oe7051.atlassian.net/browse/SSDAI-56
```

```
 SSDAI-55 : My monitor keeps flickering
```

```
 Status : To Do
```

```
 Created : 2026-06-17
```

```
 Link   : https://oe7051.atlassian.net/browse/SSDAI-55
```

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SSDAI-53 : My monitor keeps flickering all day

Status : To Do

Created : 2026-06-17

Link : <https://oe7051.atlassian.net/browse/SSDAI-53>

SSDAI-52 : Office printer on floor 3 is jammed

Status : To Do

Created : 2026-06-17

Link : <https://oe7051.atlassian.net/browse/SSDAI-52>

SSDAI-50 : Monitor flickering on laptop screen

Status : To Do

Created : 2026-06-17

Link : <https://oe7051.atlassian.net/browse/SSDAI-50>

SSDAI-49 : My keyboard is not working

Status : To Do

Created : 2026-06-17

Link : <https://oe7051.atlassian.net/browse/SSDAI-49>

SSDAI-48 : My monitor keeps flickering

Status : To Do

Created : 2026-06-17

Link : <https://oe7051.atlassian.net/browse/SSDAI-48>

john.smith — Status: TICKETS_FOUND | Count: 3

SSDAI-58 : Question about maternity leave entitlement

Status : To Do

Created : 2026-06-17

Link : <https://oe7051.atlassian.net/browse/SSDAI-58>

SSDAI-54 : How do I get an office parking permit?

Status : To Do

Created : 2026-06-17

Link : <https://oe7051.atlassian.net/browse/SSDAI-54>

SSDAI-51 : Question about maternity leave entitlement

Status : To Do

Created : 2026-06-17

Link : <https://oe7051.atlassian.net/browse/SSDAI-51>

MANUAL TICKET CREATION TEST SUMMARY

Tickets created : 3 of 3

Tickets failed : 0 of 3

Created ticket IDs: SSDAI-57, SSDAI-58, SSDAI-59

All tickets created successfully! 

Edge Case Testing: test_edge_cases.py

```
(venv) PS C:\Temp\! 0 Coursers\z_github-projects\SmartDesk_AI> python tests/test_edge_cases.py
```

```
=====
```

SmartDesk AI — Edge Case Testing

```
=====
```

Rule: Agent must NEVER crash.

Rule: Agent must ALWAYS return a string response.

Rule: Agent must ALWAYS be polite.

CATEGORY 1 — Input Boundary Tests

TEST 1 — Empty string

Input : ""

Response : It looks like you did not type anything. Please type your question and I will be...

No crash : PASS 

Content check : PASS 

TEST 1 : PASS 

```
-----
```

TEST 2 — Single space

Input : ' '

Response : It looks like you did not type anything. Please type your question and I will be...

No crash : PASS 

Content check : PASS 

TEST 2 : PASS 

```
-----
```

TEST 3 — Single character

Input : 'x'

Response : I need a bit more information to help you. Could you please describe what you ne...

No crash : PASS 

Content check : PASS 

TEST 3 : PASS 

```
-----
```

TEST 4 — Very long message — 2000 characters

Input : 'Tell me about the IT policy. Tell me about the IT policy. Te...'

Response : Your message is quite long! I have read the first part. Could you summarise your...

No crash : PASS 

Content check : PASS 

TEST 4 : PASS 

```
-----
```

TEST 5 — Message with only numbers

Input : '12345678901234567890'

Response : I am sorry I could not find information about that in my knowledge base. I can ...

No crash : PASS 

Content check : PASS 

TEST 5 : PASS 

```
-----
```

TEST 6 — Message with only special characters

Input : '!@#\$%^&*()_+-=[]{}|;:'.<>?/'

Response : I am sorry I could not find information about that in my knowledge base. I can ...

No crash : PASS 

Content check : PASS 

TEST 6 : PASS 

CATEGORY 2 — Content Edge Cases

TEST 7 — Question in all capitals

Input : 'HOW DO I RESET MY PASSWORD?'

Response : To reset your password, please follow these steps: 1. Go to [https://passwords....

No crash : PASS 

Content check : PASS 

TEST 7 : PASS 

TEST 8 — Question in all lowercase

Input : 'how do i reset my password'

Response : To reset your password, please follow these steps: 1. Go to [https://passwords....

No crash : PASS 

Content check : PASS 

TEST 8 : PASS 

TEST 9 — Question with extra whitespace

Input : ' How do I reset my password? '

Response : To reset your password, please follow these steps: 1. Go to [https://passwords....

No crash : PASS 

Content check : PASS 

TEST 9 : PASS 

TEST 10 — Repeated question marks

Input : 'How do I reset my password???'

Response : To reset your password, please follow these steps: 1. Go to [https://passwords....

No crash : PASS 

Content check : PASS 

TEST 10 : PASS 

TEST 11 — Question with emoji

Input : 'How do I reset my password 🗝?'

Response : To reset your password, please follow these steps: 1. Go to [https://passwords....

No crash : PASS 

Content check : PASS 

TEST 11 : PASS 

TEST 12 — Sensitive looking input

Input : 'My password is SuperSecret123!'

Response : I am sorry I could not find information about that in my knowledge base. I can ...

No crash : PASS 

Content check : PASS 

TEST 12 : PASS 

CATEGORY 3 — Social Edge Cases

TEST 13 — Rude frustrated message

Input : 'This system is useless and nothing ever works!'

Response : I am sorry I could not find information about that in my knowledge base. I can ...

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No crash : PASS 
Content check : PASS 
TEST 13 : PASS 

TEST 14 — Very polite formal message

Input : 'Good afternoon. I would be most grateful if you could kindly...'
Response : I am sorry I could not find information about that in my knowledge base. I can ...
No crash : PASS 
Content check : PASS 
TEST 14 : PASS 

TEST 15 — Completely off topic — weather

Input : 'What is the weather like today in San Antonio?'
Response : I am a workplace support assistant and can only help with IT support or HR polic...
No crash : PASS 
Content check : PASS 
TEST 15 : PASS 

TEST 16 — Completely off topic — personal

Input : 'Can you recommend a good restaurant near the office?'
Response : I am sorry I could not find information about that in my knowledge base. I can ...
No crash : PASS 
Content check : PASS 
TEST 16 : PASS 

TEST 17 — Asking agent its name

Input : 'What is your name?'
Response : I am sorry I could not find information about that in my knowledge base. I can ...
No crash : PASS 
Content check : PASS 
TEST 17 : PASS 

TEST 18 — Asking agent if it is AI

Input : 'Are you a real person or a chatbot?'
Response : I am sorry I could not find information about that in my knowledge base. I can ...
No crash : PASS 
Content check : PASS 
TEST 18 : PASS 

CATEGORY 4 — Session State Edge Cases

TEST 19 — Invalid email format when email expected

Input : 'notanemail'
Response : That does not look like a valid email address. Please enter your work email add...
No crash : PASS 
Content check : PASS 
TEST 19 : PASS 

TEST 20 — Random text when yes/no expected

Input : 'purple elephant'
Response : I would like to create a support ticket for you. Please review the details belo...

17 SMARTDESK_AI Documented Test Results

No crash : PASS 

Content check : PASS 

TEST 20 : PASS 

CATEGORY 5 — Repetition Tests

TEST 21 — Same question asked 3 times in same session

Attempt 1 : To reset your password, please follow these steps: 1. Go to...

Attempt 2 : To reset your password, please follow these steps: 1. Go to...

Attempt 3 : To reset your password, please follow these steps: 1. Go to...

All 3 attempts responded : PASS 

TEST 21 : PASS 

EDGE CASE TEST SUMMARY

Total tests : 21

Tests passed : 21

Tests failed : 0

Crashes : 0

Zero crashes! 

All edge case tests passed! 

Error Handling Test Summary: test_error_handling.py

```
(venv) PS C:\Temp\! 0 Coursers\z_github-projects\SmartDesk_AI> python tests/test_error_handling.py
```

```
=====
SmartDesk AI — Error Handling Test
=====
```

```
TEST 1 — API error response builder
```

```
Response preview : I am sorry — I am having trouble connecting to OpenAI right now.
```

```
This is usuall...
```

```
TEST 1 : PASS 
```

```
-----
TEST 2 — Jira ticket creation API failure
```

```
[Agent] Jira ticket creation error: Simulated Jira connection timeout
```

```
Response preview : I am sorry — I am having trouble connecting to Jira right now.
```

```
This is usually ...
```

```
TEST 2 : PASS 
```

```
-----
TEST 3 — Jira ticket status API failure
```

```
[Agent] Jira status lookup error: Simulated Jira authentication failure
```

```
Response preview : I am sorry — I am having trouble connecting to Jira right now.
```

```
This is usually ...
```

```
TEST 3 : PASS 
```

```
-----
TEST 4 — RAG pipeline OpenAI API failure
```

```
[Agent] RAG error: Simulated OpenAI rate limit exceeded
```

```
Response preview : I am sorry — I am having trouble connecting to the knowledge base right now.
```

```
Th...
```

```
TEST 4 : PASS 
```

```
-----
TEST 5 — Agent never crashes on unexpected exception
```

```
[Agent] Intent detection error: Simulated catastrophic failure
```

```
Response preview : I am sorry I could not find information about that in my knowledge base.
```

```
I can ...
```

```
TEST 5 : PASS 
```

```
=====
ERROR HANDLING TEST SUMMARY
=====
```

```
Tests passed : 5 of 5
```

```
Tests failed : 0 of 5
```

```
All tests passed! 
```

```
Your agent handles all API failures gracefully.
```

Knowledge Base Answer Verification: test_flow_a.py

(venv) PS C:\Temp\! 0 Coursers\z_github-projects\SmartDesk_AI> python tests/test_flow_a.py

SmartDesk AI — Flow A Official Test
Knowledge Base Answer Verification

Testing that the agent correctly answers questions
from the knowledge base without hallucinating.

TEST 1 — IT — Password Reset

Question : How do I reset my password?

Response : To reset your password, please follow these steps: 1. Go to [https://passwords.roadmapconsulting.com]. 2. Select "Forgot Password" or "Reset Password." 3. Enter your Roadmap Consulting username or wo...

Contains required words : PASS 

No escalation words : PASS 

TEST 1 : PASS 

TEST 2 — IT — VPN Setup

Question : How do I set up VPN on my Windows laptop?

Response : To set up the VPN on your Windows laptop, please follow these steps: 1. Sign in to your Windows computer. 2. Connect to the internet using Wi-Fi or a wired connection. 3. Open Microsoft Edge, Chrome,...

Contains required words : PASS 

No escalation words : PASS 

TEST 2 : PASS 

TEST 3 — IT — MFA Setup

Question : How do I set up MFA on my phone?

Response : To set up MFA on your phone, please follow these steps: 1. On your computer, sign in to your Roadmap Consulting account. 2. When prompted to set up MFA, select **Begin Setup** or **Next**. 3. Choose ...

Contains required words : PASS 

No escalation words : PASS 

TEST 3 : PASS 

TEST 4 — HR — Casual Leave

Question : How many casual leave days do I get per year?

Response : Eligible full-time employees receive 8 Casual Leave days per calendar year. Casual Leave may be taken as a full day, half day, or hourly increment if approved by the manager and supported by the Leave...

Contains required words : PASS 

No escalation words : PASS 

TEST 4 : PASS 

TEST 5 — HR — Work From Home

Question : What is the work from home policy?

Response : The work from home policy at Roadmap Consulting supports flexible work arrangements where business needs, role requirements, performance expectations, customer commitments, and security requirements c...

Contains required words : PASS 

No escalation words : PASS 

TEST 5 : PASS 

17 SMARTDESK_AI Documented Test Results

BONUS CHECK — Direct RAG chain verification

Confirming RAG returns ANSWER_FOUND for KB questions

'How do I reset my password?' → ANSWER_FOUND 

'How many sick leave days do I get?' → ANSWER_FOUND 

'How do I set up MFA?' → ANSWER_FOUND 

RAG direct checks : 3 of 3 passed

=====

FLOW A TEST SUMMARY

=====

Agent tests passed : 5 of 5

Agent tests failed : 0 of 5

RAG direct checks : 3 of 3

All Flow A tests passed! 

Escalation to Ticket Creation Verification: test_flow_b.py

(venv) PS C:\Temp\! 0 Coursers\z_github-projects\SmartDesk_AI> python tests/test_flow_b.py

SmartDesk AI — Flow B Official Test

Escalation to Ticket Creation Verification, Creates Jira support tickets when it does not have an answer

Testing that the agent correctly escalates
out-of-scope questions to Jira ticket creation.

SCENARIO 1 — IT out of scope — monitor flickering

Question : My monitor keeps flickering all day

Email : jane.doe@roadmapconsulting.com

Confirm : YES

Turn 1 response : I am sorry I could not find information about that in my knowledge base. I can create a support ticket...

Turn 2 response : I would like to create a support ticket for you. Please review the details below: -----...

Ticket created : SSDAI-60 

Turn 3 response : Your support ticket has been created successfully! Ticket ID : SSDAI-60 Title : My monito...

Turn 1 - Escalation offered : PASS 

Turn 2 - Ticket summary shown : PASS 

Turn 3 - Ticket created : PASS 

SCENARIO 1 : PASS 

SCENARIO 2 — Facilities out of scope — office parking

Question : How do I get an office parking permit?

Email : john.smith@roadmapconsulting.com

Confirm : YES

Turn 1 response : I am sorry I could not find information about that in my knowledge base. I can create a support ticket...

Turn 2 response : I would like to create a support ticket for you. Please review the details below: -----...

Ticket created : SSDAI-61 

Turn 3 response : Your support ticket has been created successfully! Ticket ID : SSDAI-61 Title : How do I ...

Turn 1 - Escalation offered : PASS 

Turn 2 - Ticket summary shown : PASS 

Turn 3 - Ticket created : PASS 

SCENARIO 2 : PASS 

SCENARIO 3 — Employee cancels ticket creation

Question : The printer on floor 3 is jammed again

Email : jane.doe@roadmapconsulting.com

Confirm : NO

Turn 1 response : I am sorry I could not find information about that in my knowledge base. I can create a support ticket...

Turn 2 response : I would like to create a support ticket for you. Please review the details below: -----...

Turn 3 response : No problem! The ticket has not been created. Is there anything else I can help you with?...

Turn 1 - Escalation offered : PASS 

Turn 2 - Ticket summary shown : PASS 

Turn 3 - Ticket cancelled : PASS 

SCENARIO 3 : PASS 

SCENARIO 4 — Verify created tickets exist in Jira

Checking 4 ticket(s) in Jira...

SSDAI-60 : FOUND in Jira 

Summary : My monitor keeps flickering all day

Status : To Do

SSDAI-60 : FOUND in Jira 

Summary : My monitor keeps flickering all day

Status : To Do

SSDAI-61 : FOUND in Jira 

Summary : How do I get an office parking permit?

Status : To Do

SSDAI-61 : FOUND in Jira 

Summary : How do I get an office parking permit?

Status : To Do

SCENARIO 4 : PASS 

=====

FLOW B TEST SUMMARY

=====

Scenarios passed : 4 of 4

Scenarios failed : 0 of 4

Tickets created : ['SSDAI-60', 'SSDAI-60', 'SSDAI-61', 'SSDAI-61']

All Flow B tests passed! 

Ticket Status Check Verification: test_flow_c.py

```
(venv) PS C:\Temp\! 0 Coursers\z_github-projects\SmartDesk_AI> python tests/test_flow_c.py
```

```
=====
```

SmartDesk AI — Flow C Official Test

Ticket Status Check Verification

Provides Jira ticket status by email address (key)

```
=====
```

Testing that the agent correctly retrieves and displays ticket status for employees.

SCENARIO 1 — Employee with multiple tickets

Question : What is the status of my tickets?

Email : jane.doe@roadmapconsulting.com

Expects : Tickets found

Turn 1 response : Could you please share your work email address so I can look that up for you?...

Turn 2 response : I found 10 ticket(s) for jane.doe@roadmapconsulting.com: 📄 SSDAI-60 : My monitor keeps flickering all day 📅 Status : To Do 📅 Created : 2026-06...

Turn 1 - Status intent detected : PASS ✓

Turn 2 - Tickets displayed : PASS ✓

Session email remembered : PASS ✓

SCENARIO 1 : PASS ✓

```
-----
```

SCENARIO 2 — Employee with single ticket

Question : Any updates on my support request?

Email : john.smith@roadmapconsulting.com

Expects : Tickets found

Turn 1 response : Could you please share your work email address so I can look that up for you?...

Turn 2 response : I found 4 ticket(s) for john.smith@roadmapconsulting.com: 📄 SSDAI-61 : How do I get an office parking permit? 📅 Status : To Do 📅 Created : 202...

Turn 1 - Status intent detected : PASS ✓

Turn 2 - Tickets displayed : PASS ✓

Session email remembered : PASS ✓

SCENARIO 2 : PASS ✓

```
-----
```

SCENARIO 3 — Employee with no tickets

Question : Can you check my ticket status please?

Email : new.employee@roadmapconsulting.com

Expects : No tickets

Turn 1 response : Could you please share your work email address so I can look that up for you?...

Turn 2 response : I could not find any open tickets for new.employee@roadmapconsulting.com. Would you like me to create a new support ticket? If so please describe the...

Turn 1 - Status intent detected : PASS ✓

Turn 2 - No tickets message : PASS ✓

Session email remembered : PASS ✓

SCENARIO 3 : PASS ✓

```
-----
```

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SCENARIO 4 — Session memory across multiple turns

Testing that email is remembered after first lookup

Turn 3 response : I found 10 ticket(s) for jane.doe@roadmapconsulting.com: 🗑️ SSDAI-60 : My monitor keeps flickering all day 📅 Status : To Do 📅 Created : 2026-06...

Email not asked again : PASS ✅

Tickets shown directly : PASS ✅

SCENARIO 4 : PASS ✅

SCENARIO 5 — Direct Jira API verification

Confirming get_ticket_status works independently

jane.doe@roadmapconsulting.com : 10 ticket(s) ✅

john.smith@roadmapconsulting.com : 4 ticket(s) ✅

SCENARIO 5 : PASS ✅

=====

FLOW C TEST SUMMARY

=====

Scenarios passed : 5 of 5

Scenarios failed : 0 of 5

All Flow C tests passed! ✅

Ticket Status Comprehensive Test: test_get_status.py

(venv) PS C:\Temp\! 0 Coursers\z_github-projects\SmartDesk_AI> python tests/test_get_status.py

=====

SmartDesk AI — Ticket Status Comprehensive Test

=====

TEST 1 — Look up tickets for email that HAS tickets

Status : TICKETS_FOUND

Count : 6

Reason : Found 6 ticket(s) for demo@roadmapconsulting.com

Formatted output:

📌 SSDAI-47 : My monitor keeps flickering

📌 Status : To Do

📅 Created : 2026-06-17

🔗 Link : <https://oe7051.atlassian.net/browse/SSDAI-47>

📌 SSDAI-43 : My Keyboard makes a clicking noise

📌 Status : To Do

📅 Created : 2026-06-12

💬 Update : Thank you for your inquiry. Please know a support specialist has been assigned and will provide the

🔗 Link : <https://oe7051.atlassian.net/browse/SSDAI-43>

📌 SSDAI-42 : My printer stopped working

📌 Status : To Do

📅 Created : 2026-06-12

🔗 Link : <https://oe7051.atlassian.net/browse/SSDAI-42>

📌 SSDAI-41 : I can't connect to the internet

📌 Status : To Do

📅 Created : 2026-06-12

🔗 Link : <https://oe7051.atlassian.net/browse/SSDAI-41>

📌 SSDAI-40 : VPN is not working

📌 Status : To Do

📅 Created : 2026-06-12

💬 Update : Your request is in the queue

🔗 Link : <https://oe7051.atlassian.net/browse/SSDAI-40>

📌 SSDAI-39 : My mouse won't work

📌 Status : To Do

📅 Created : 2026-06-11

🔗 Link : <https://oe7051.atlassian.net/browse/SSDAI-39>

TEST 1 : PASS ✓

TEST 2 — Look up tickets for email with NO tickets

Status : TICKETS_NONE

Count : 0

Reason : No tickets found for nobody.here@roadmapconsulting.com

TEST 2 : PASS ✓

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TEST 3 — Look up a specific ticket by ID (SSDAI-47)

Status : TICKETS_FOUND

Ticket ID : SSDAI-47

Summary : My monitor keeps flickering

Ticket Status: To Do

Created : 2026-06-17

URL : <https://oe7051.atlassian.net/browse/SSDAI-47>

TEST 3 : PASS 

TEST 4 — Look up a ticket ID that does NOT exist (SSDAI-99999)

Status : TICKETS_FAILED

Reason : Jira error: 404 - Issue does not exist or you do not have permission to see it.

TEST 4 : PASS 

=====

TICKET STATUS TEST SUMMARY

=====

Tests passed : 4 of 4

Tests failed : 0 of 4

All tests passed! 

Jira integration is fully working.

Graceful Handling Test: test_graceful_handling.py

```
(venv) PS C:\Temp\! 0 Coursers\z_github-projects\SmartDesk_AI> python tests/test_graceful_handling.py
```

```
=====
```

SmartDesk AI — Graceful Handling Test

```
=====
```

TEST 1 — Empty message

Input : "

Response : It looks like you did not type anything. Please type your question and I will be happy to help!...

TEST 1 : PASS 

```
-----
```

TEST 2 — Single character input

Input : 'x'

Response : I need a bit more information to help you. Could you please describe what you need?...

TEST 2 : PASS 

```
-----
```

TEST 3 — Very long message over 1000 characters

Input : 'This is a very long message This is a very long message This'

Response : Your message is quite long! I have read the first part. Could you summarise your question in one or ...

TEST 3 : PASS 

```
-----
```

TEST 4 — Gibberish keyboard mashing

Input : 'qwrt zxcvbnm plkjhg'

Response : I am sorry I could not find information about that in my knowledge base. I can create a support tic...

TEST 4 : PASS 

```
-----
```

TEST 5 — Simple greeting

Input : 'Hello'

Response : Hello! Great to hear from you. I am SmartDesk AI, your IT and HR support assistant at Roadmap Consu...

TEST 5 : PASS 

```
-----
```

TEST 6 — Good morning greeting

Input : 'Good morning'

Response : Hello! Great to hear from you. I am SmartDesk AI, your IT and HR support assistant at Roadmap Consu...

TEST 6 : PASS 

```
-----
```

TEST 7 — Thank you message

Input : 'Thanks that helped a lot'

Response : You are very welcome! I am glad I could help. Is there anything else I can assist you with today?...

TEST 7 : PASS 

```
-----
```

TEST 8 — Completely off topic question

Input : 'What is the capital of France?'

Response : I am sorry I could not find information about that in my knowledge base. I can create a support tic...

TEST 8 : PASS 

```
-----
```

TEST 9 — Out of scope topic from gaps list

Input : 'My monitor keeps flickering'

Response : I am sorry I could not find information about that in my knowledge base. I can create a support tic...

TEST 9 : PASS 

```
-----
```

17 SMARTDESK_AI Documented Test Results

TEST 10 — Frustrated employee message

Input : 'This is useless nothing works!'

Response : I am sorry I could not find information about that in my knowledge base. I can create a support tic...

TEST 10 : PASS 

=====

GRACEFUL HANDLING TEST SUMMARY

=====

Tests passed : 10 of 10

Tests failed : 0 of 10

All tests passed! 

Your agent handles all edge cases gracefully.

Jira Live Connection Test: test_jira_connection.py

(venv) PS C:\Temp\! 0 Coursers\z_github-projects\SmartDesk_AI> python tests/test_jira_connection.py

```
=====
SmartDesk AI — Jira Live Connection Test
=====
```

Test 1: Connecting to Jira...

Server : https://oe7051.atlassian.net

Email : oe7051@gmail.com

Connection : SUCCESS 

Test 2: Getting your account details...

Display Name : Oscar Espinoza 

Email Address : oe7051@gmail.com 

Account ID : 712020:d2ac8c7a-... 

Test 3: Looking for project 'SSDAI'...

Project Name : Scrum-SmartDeskAI 

Project Key : SSDAI 

Project ID : 10004 

Test 4: Getting available issue types...

Issue types found: ['Epic', 'Subtask', 'Task', 'Support', 'Service Request', 'Incident'] 

Test 5: Counting existing tickets in SSDAI...

Tickets found : 13 

Most recent tickets:

SSDAI-54 : How do I get an office parking permit?

SSDAI-53 : My monitor keeps flickering all day

SSDAI-52 : Office printer on floor 3 is jammed

```
=====
JIRA CONNECTION TEST COMPLETE
=====
```

If all 5 tests passed your Jira connection is fully working.

LangGraph Installation Test: test_langgraph.py

(venv) PS C:\Temp\! 0 Coursers\z_github-projects\SmartDesk_AI> python tests/test_langgraph.py

SmartDesk AI — LangGraph Installation Test

Test 1: Importing core LangGraph...

StateGraph : OK 

END : OK 

Test 2: Importing TypedDict for state management...

TypedDict : OK 

Annotated : OK 

List : OK 

Test 3: Importing LangChain message types...

HumanMessage : OK 

AIMessage : OK 

SystemMessage : OK 

Test 4: Building a minimal test graph...

Graph built : OK 

Graph compiled : OK 

Graph ran : OK 

Test output : HELLO SMARTDESK 

Test 5: Confirming all agent dependencies are ready...

rag_chain : OK 

jira_tools : OK 

rag_config : OK 

LANGGRAPH INSTALLATION TEST SUMMARY

Tests passed : 5 of 5

Tests failed : 0 of 5

All tests passed! 

LangGraph is installed and all agent dependencies are ready.

RAG Chain Comprehensive Test: test_rag_chain.py

(venv) PS C:\Temp\! 0 Coursers\z_github-projects\SmartDesk_AI> python tests/test_rag_chain.py

SmartDesk AI — RAG Chain Comprehensive Test

TEST 1 — IN-SCOPE IT

Query : How do I set up VPN on my Windows laptop?

Expected : ANSWER_FOUND

Got : ANSWER_FOUND

Sources : ['knowledge_base/it_qa.json', 'knowledge_base/it_support_guide.md']

Verdict : PASS 

Answer Preview:

To set up the VPN on your Windows laptop, please follow these steps:

1. Sign in to your Windows computer.
2. Connect to the internet using Wi-Fi or a wired connection.
3. Open Microsoft Edge, Chrome, or another approved browser.
4. Go to the IT Portal: [https://it.roadmapconsulting.com](https://it....)

TEST 2 — IN-SCOPE HR

Query : How many sick leave days do I get per year?

Expected : ANSWER_FOUND

Got : ANSWER_FOUND

Sources : ['knowledge_base/hr_qa.json']

Verdict : PASS 

Answer Preview:

Eligible full-time employees receive 10 Sick Leave days per calendar year. Part-time employees may receive Sick Leave on a prorated basis based on their regular work schedule.

Is there anything else I can help with?...

TEST 3 — OUT-OF-SCOPE

Query : My monitor keeps flickering all day

Expected : ANSWER_NOT_FOUND

Got : ANSWER_NOT_FOUND

Sources : []

Verdict : PASS 

Answer Preview:

I don't have enough information about that in my knowledge base....

RAG CHAIN TEST SUMMARY

Tests passed : 3 of 3

Tests failed : 0 of 3

All tests passed!

RAG chain is working correctly.

Retrieval Pipeline Test: test_retrieval.py

```
(venv) PS C:\Temp\! 0 Coursers\z_github-projects\SmartDesk_AI> python tests/test_retrieval.py
```

```
=====
```

SmartDesk AI — Retrieval Pipeline Test

```
=====
```

Connecting to ChromaDB...

C:\Temp\! 0 Coursers\z_github-projects\SmartDesk_AI\tests\test_retrieval.py:28: LangChainDeprecationWarning: The class `Chroma` was deprecated in LangChain 0.2.9 and will be removed in 1.0. An updated version of the class exists in the `langchain-chroma` package and should be used instead. To use it run `pip install -U `langchain-chroma` and import as `from `langchain_chroma import Chroma``.

```
db = Chroma(
Connected! Total chunks in database: 175
```

TEST 1 — IT Support

Query : How do I reset my password?

Expects : password reset portal or steps

Found 3 matching chunks:

Match 1:

Score : 31.9% similar

Source : knowledge_base/it_qa.json

Preview : Question: How do I reset my password if I forgot it? Answer: Go to [https://passwords.roadmapconsulting.com] and select Forgot Password or Reset Password. Enter your Roadmap Consulting username or wor...

Match 2:

Score : 15.4% similar

Source : knowledge_base/it_support_guide.md

Preview : * You may be asked to verify using MFA. * You may receive a verification code. * You may need to answer security questions if configured. 6. Enter a new password that meets Roadmap Consulting pa...

Match 3:

Score : 9.6% similar

Source : knowledge_base/it_support_guide.md

Preview : --- # 1. Password Reset Your Roadmap Consulting password protects access to your computer, email, VPN, business applications, and company data. Never share your password with anyone, including cowor...

TEST 2 — IT Support

Query : How do I set up VPN on Windows?

Expects : VPN installation steps

Found 3 matching chunks:

Match 1:

Score : 35.4% similar

Source : knowledge_base/it_qa.json

Preview : Question: How do I install the VPN client on my Windows 11 laptop? Answer: Go to [https://it.roadmapconsulting.com], search for VPN Client for Windows 11, and download the approved Windows VPN install...

Match 2:

Score : 31.8% similar

Source : knowledge_base/it_support_guide.md

Preview : 1. Sign in to your Windows 11 computer. 2. Connect to the internet using Wi-Fi or a wired connection. 3. Open Microsoft Edge, Chrome, or another approved browser. 4. Go to the IT Portal: [https://it.r...

Match 3:

Score : 21.5% similar

Source : knowledge_base/it_qa.json

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Preview : Question: How do I install the VPN client on my Mac? Answer: Go to [https://it.roadmapconsulting.com], search for VPN Client for Mac, and download the approved Mac VPN installer. Open the downloaded

TEST 3 — HR Policy

Query : How many casual leave days do I get?

Expects : number of casual leave days

Found 3 matching chunks:

Match 1:

Score : 60.4% similar

Source : knowledge_base/hr_qa.json

Preview : Question: How many Casual Leave days do I get each year? Answer: Eligible full-time employees receive 8 Casual Leave days per calendar year. Casual Leave may be taken as a full day, half day, or hourl...

Match 2:

Score : 37.6% similar

Source : knowledge_base/hr_leave_policy.md

Preview : For questions about legal eligibility, special medical situations, pregnancy-related needs, disability accommodation, or extended leave, contact HR at **[hr@roadmapconsulting.com]** or **ext. 2525**: ...

Match 3:

Score : 33.2% similar

Source : knowledge_base/hr_leave_policy.md

Preview : * Full day * Half day * Hourly leave, if available 6. Enter a brief reason for the request. 7. Confirm your available Casual Leave balance. 8. Submit the request. 9. Wait for manager approval be...

TEST 4 — HR Policy

Query : What is the work from home policy?

Expects : work from home eligibility or rules

Found 3 matching chunks:

Match 1:

Score : 28.8% similar

Source : knowledge_base/hr_leave_policy.md

Preview : Work from home, also called WFH, is a work arrangement. It is not a leave type. Employees working from home are expected to be available, responsive, productive, and working during approved work hours...

Match 2:

Score : 21.9% similar

Source : knowledge_base/hr_leave_policy.md

Preview : ### Encashment Request Steps 1. Log in to https://hr.roadmapconsulting.com. 2. Open an HR request. 3. Select **Leave Encashment Inquiry** or the closest available ...

Match 3:

Score : 21.4% similar

Source : knowledge_base/hr_qa.json

Preview : Question: Can I work from home, and who is eligible? Answer: Work from home eligibility depends on role, performance, work responsibilities, security needs, customer commitments, and manager approval...

TEST 5 — IT Support

Query : How do I set up MFA on my phone?

Expects : MFA setup steps

Found 3 matching chunks:

Match 1:

Score : 56.0% similar

Source : knowledge_base/it_support_guide.md

17 SMARTDESK_AI Documented Test Results

Preview : --- ## 3.2 How to Set Up MFA on a Smartphone Use this process when registering a phone for MFA.

Steps 1. On your computer, sign in to your Roadmap Consulting account. 2. When prompted to set u...

Match 2:

Score : 52.8% similar

Source : knowledge_base/it_support_guide.md

Preview : * Text message code. * Phone call. * Push notification through an authenticator app. 6. Follow the on-screen instructions. 7. If you receive a text message code, enter the code on your computer....

Match 3:

Score : 51.3% similar

Source : knowledge_base/it_qa.json

Preview : Question: How do I set up MFA on my smartphone? Answer: Sign in to your Roadmap Consulting account on your computer and choose Mobile Phone or Smartphone when prompted for MFA setup. Enter your mobile...

=====

RETRIEVAL TEST COMPLETE

=====

What to look for in the results above:

Good score : 40% or higher similarity

Good source : Points to the right KB file

Good preview : Shows relevant content for the query

Security Scan (secrets) : test_security.py

```
(venv) PS C:\Temp\! 0 Coursers\z_github-projects\SmartDesk_AI> python tests/test_security.py
```

```
=====
```

SmartDesk AI — Security Scan

No Hard-Coded Secrets Verification

```
=====
```

Scanning 82 files...

Python files : 37

Markdown files: 45

Checking: OpenAI API Key

Pattern: sk-p****

Not found in any file : PASS 

Checking: Jira API Token

Pattern: ATAT****

Not found in any file : PASS 

Checking: Jira Email

Pattern: oe70****

Not found in any file : PASS 

Checking: .env is in .gitignore

.env in .gitignore : PASS 

Checking: .env file is not tracked by git

.env not tracked : PASS 

Checking: chroma_db is in .gitignore

chroma_db in .gitignore: PASS 

Checking: venv is in .gitignore

venv in .gitignore : PASS 

Checking: Python files use os.getenv for secrets

No suspicious patterns : PASS 

```
=====
```

SECURITY SCAN SUMMARY

```
=====
```

Checks passed : 8

Checks failed : 0

Warnings : 0

All security checks passed! 

No secrets found. Your project is safe to submit.

Confidence Threshold Test: test_threshold.py

```
(venv) PS C:\Temp\! 0 Coursers\z_github-projects\SmartDesk_AI> python tests/test_threshold.py
```

```
=====
```

SmartDesk AI — Confidence Threshold Test

Threshold set at: 15.0%

```
=====
```

Test 1 — IN-SCOPE

Query : How do I reset my password?

Expected : FOUND

Got : FOUND

Scores : [31.9, 15.4, 9.6]

Reason : Best match score 31.9% is above threshold of 15.0%.

Verdict : PASS 

```
-----
```

Test 2 — IN-SCOPE

Query : How many sick leave days do I get?

Expected : FOUND

Got : FOUND

Scores : [41.5, 18.3, 18.1]

Reason : Best match score 41.5% is above threshold of 15.0%.

Verdict : PASS 

```
-----
```

Test 3 — IN-SCOPE

Query : How do I set up MFA on my phone?

Expected : FOUND

Got : FOUND

Scores : [56.0, 52.8, 51.3]

Reason : Best match score 56.0% is above threshold of 15.0%.

Verdict : PASS 

```
-----
```

Test 4 — OUT-OF-SCOPE

Query : My monitor keeps flickering

Expected : NOT_FOUND

Got : NOT_FOUND

Scores : [-38.8, -41.3, -42.8]

Reason : Best match score -38.8% is below threshold of 15.0%.

Verdict : PASS 

```
-----
```

Test 5 — OUT-OF-SCOPE

Query : How do I get an office parking permit?

Expected : NOT_FOUND

Got : NOT_FOUND

Scores : [-33.1, -33.7, -36.9]

Reason : Best match score -33.1% is below threshold of 15.0%.

Verdict : PASS 

```
-----
```

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Test 6 — OUT-OF-SCOPE

Query : The office printer is jammed

Expected : NOT_FOUND

Got : NOT_FOUND

Scores : [-17.3, -35.2, -35.8]

Reason : Best match score -17.3% is below threshold of 15.0%.

Verdict : PASS 

=====

THRESHOLD TEST SUMMARY

=====

Tests passed : 6 of 6

Tests failed : 0 of 6

All tests passed!

Confidence threshold is working correctly.

Manual Ticket Status Lookup Test: test_ticket_status.py

```
(venv) PS C:\Temp\! 0 Coursers\z_github-projects\SmartDesk_AI> python tests/test_ticket_status.py
```

```
=====
SmartDesk AI — Manual Ticket Status Lookup Test
=====
```

TEST 1 — Employee with MULTIPLE tickets

```
jane.doe has 2 tickets: monitor + printer
```

```
Status : TICKETS_FOUND
```

```
Count : 5 ticket(s) found
```

```
Reason : Found 5 ticket(s) for jane.doe@roadmapconsulting.com
```

SSDAI-53 : My monitor keeps flickering all day

```
Status : To Do
```

```
Created : 2026-06-17
```

```
Link : https://oe7051.atlassian.net/browse/SSDAI-53
```

SSDAI-52 : Office printer on floor 3 is jammed

```
Status : To Do
```

```
Created : 2026-06-17
```

```
Link : https://oe7051.atlassian.net/browse/SSDAI-52
```

SSDAI-50 : Monitor flickering on laptop screen

```
Status : To Do
```

```
Created : 2026-06-17
```

```
Link : https://oe7051.atlassian.net/browse/SSDAI-50
```

SSDAI-49 : My keyboard is not working

```
Status : To Do
```

```
Created : 2026-06-17
```

```
Link : https://oe7051.atlassian.net/browse/SSDAI-49
```

SSDAI-48 : My monitor keeps flickering

```
Status : To Do
```

```
Created : 2026-06-17
```

```
Link : https://oe7051.atlassian.net/browse/SSDAI-48
```

```
TEST 1 : PASS ✓
```

TEST 2 — Employee with a SINGLE ticket

```
john.smith has 1 ticket: maternity leave
```

```
Status : TICKETS_FOUND
```

```
Count : 2 ticket(s) found
```

```
Reason : Found 2 ticket(s) for john.smith@roadmapconsulting.com
```

SSDAI-54 : How do I get an office parking permit?

```
Status : To Do
```

```
Created : 2026-06-17
```

```
Link : https://oe7051.atlassian.net/browse/SSDAI-54
```

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SSDAI-51 : Question about maternity leave entitlement

Status : To Do

Created : 2026-06-17

Link : <https://oe7051.atlassian.net/browse/SSDAI-51>

TEST 2 : PASS ✓

TEST 3 — Employee with NO tickets

Status : TICKETS_NONE

Count : 0 ticket(s) found

Reason : No tickets found for new.employee@roadmapconsulting.com

TEST 3 : PASS ✓

TEST 4 — Look up specific ticket SSDAI-47 by ID

Status : TICKETS_FOUND

Ticket ID : SSDAI-47

Summary : My monitor keeps flickering

Ticket Status: To Do

Created : 2026-06-17

URL : <https://oe7051.atlassian.net/browse/SSDAI-47>

TEST 4 : PASS ✓

TEST 5 — Simulate a full Flow C agent conversation

I found 5 ticket(s) for jane.doe@roadmapconsulting.com:

SSDAI-53 : My monitor keeps flickering all day

Status : To Do

Created : 2026-06-17

Link : <https://oe7051.atlassian.net/browse/SSDAI-53>

SSDAI-52 : Office printer on floor 3 is jammed

Status : To Do

Created : 2026-06-17

Link : <https://oe7051.atlassian.net/browse/SSDAI-52>

SSDAI-50 : Monitor flickering on laptop screen

Status : To Do

Created : 2026-06-17

Link : <https://oe7051.atlassian.net/browse/SSDAI-50>

SSDAI-49 : My keyboard is not working

Status : To Do

Created : 2026-06-17

Link : <https://oe7051.atlassian.net/browse/SSDAI-49>

SSDAI-48 : My monitor keeps flickering

Status : To Do

Created : 2026-06-17

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 Link : <https://oe7051.atlassian.net/browse/SSDAI-48>

Is there anything else I can help you with?

TEST 5 : PASS 

=====

MANUAL STATUS LOOKUP TEST SUMMARY

=====

Tests passed : 5 of 5

Tests failed : 0 of 5

All tests passed! 

Evaluator Poise (confidence) rubric check: test_upoise_checks.py

(venv) PS C:\Temp\! 0 Coursers\z_github-projects\SmartDesk_AI> python tests/test_upoise_checks.py

=====

SmartDesk AI — Evaluator Rubric Poise (Confidence) Check
(Updated for src/ subfolder structure)

=====

CATEGORY A — Knowledge Base & RAG Pipeline [25 marks]

A1: Quality and breadth of KB content

- ✓ PASS [1pts] IT support guide exists and is substantial
- ✓ PASS [1pts] HR leave policy exists and is substantial
- ✓ PASS [1pts] External HR dataset included
- ✓ PASS [1pts] Deliberate gaps documented
- ✓ PASS [1pts] 30+ Q&A pairs total (found 41)

A2: Chunking strategy and justification

- ✓ PASS [1pts] CHUNK_SIZE defined in rag_config.py
- ✓ PASS [1pts] CHUNK_OVERLAP defined in rag_config.py
- ✓ PASS [1pts] Embedding model configured
- ✓ PASS [1pts] Chunking strategy documented
- ✓ PASS [1pts] rag_config.py exists and centralises settings

A3: Retrieval quality

- ✓ PASS [2pts] retrieval_with_threshold.py exists
- ✓ PASS [2pts] ChromaDB vector store exists locally
- ✓ PASS [1pts] test_retrieval.py exists

A4: LLM grounding — no hallucination

- ✓ PASS [2pts] SYSTEM_PROMPT in rag_chain.py
- ✓ PASS [1pts] Context injection in prompt
- ✓ PASS [2pts] Hallucination guard phrase present

A5: Confidence / escalation logic

- ✓ PASS [2pts] Confidence threshold check implemented
- ✓ PASS [2pts] RESULT_FOUND and RESULT_NOT_FOUND defined
- ✓ PASS [1pts] test_threshold.py exists

CATEGORY A SCORE: 25 / 25

CATEGORY B — Ticket Creation Write Operation [20 marks]

B1: Successful external API integration

- ✓ PASS [2pts] jira library imported
- ✓ PASS [2pts] create_ticket function defined
- ✓ PASS [1pts] Jira client connection established

B2: All mandatory fields collected

- ✓ PASS [1pts] Summary field
- ✓ PASS [1pts] Description field

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- ✓ PASS [1pts] Employee email
- ✓ PASS [1pts] Category field
- ✓ PASS [1pts] Issue type

B3: Human-in-the-loop confirmation

- ✓ PASS [2pts] awaiting_confirmation in session
- ✓ PASS [1pts] Ticket summary shown before confirm
- ✓ PASS [1pts] YES path creates ticket
- ✓ PASS [1pts] NO path cancels politely

B4: Post-creation feedback with ticket ID/link

- ✓ PASS [2pts] Ticket ID returned
- ✓ PASS [2pts] Ticket URL returned
- ✓ PASS [1pts] TICKET_CREATED status returned

CATEGORY B SCORE: 20 / 20

CATEGORY C — Ticket Status Check Read Operation [15 marks]

C1: Status retrieval via API

- ✓ PASS [2pts] get_ticket_status function defined
- ✓ PASS [1pts] get_ticket_by_id function defined
- ✓ PASS [1pts] JQL search by email
- ✓ PASS [1pts] Status field retrieved

C2: Multiple-ticket handling

- ✓ PASS [2pts] format_tickets helper function
- ✓ PASS [1pts] Multiple tickets returned as list
- ✓ PASS [1pts] Ticket count returned
- ✓ PASS [1pts] maxResults limits query

C3: Graceful handling when no tickets found

- ✓ PASS [2pts] TICKETS_NONE status returned
- ✓ PASS [2pts] Empty results handled gracefully
- ✓ PASS [1pts] test_flow_c.py exists

CATEGORY C SCORE: 15 / 15

CATEGORY D — Agent Orchestration & Routing [15 marks]

D1: Correct intent detection and routing

- ✓ PASS [2pts] detect_intent function defined
- ✓ PASS [1pts] KB_QUERY and CHECK_STATUS intents defined
- ✓ PASS [1pts] Intent system prompt defined
- ✓ PASS [1pts] Routing to correct handler functions

D2: Smooth multi-turn conversations

- ✓ PASS [1pts] create_session or session management
- ✓ PASS [1pts] Employee email stored in session
- ✓ PASS [1pts] Chat history maintained

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- ✓ PASS [1pts] awaiting_email state tracking
- ✓ PASS [1pts] Workflow files separate flows A B C

D3: Out-of-scope and ambiguous query handling

- ✓ PASS [1pts] UNCLEAR intent handled
- ✓ PASS [1pts] Unclear response function
- ✓ PASS [2pts] Out of scope escalates to ticket
- ✓ PASS [1pts] validate_message or input validation

CATEGORY D SCORE: 15 / 15

CATEGORY E — Code Quality & Documentation [15 marks]

E1: Clean, modular, readable code

- ✓ PASS [2pts] All 6 core modules exist in src/
- ✓ PASS [1pts] Functions have docstrings (40 found)
- ✓ PASS [2pts] 15+ test files in tests/ folder (17 found)

E2: README with setup instructions

- ✓ PASS [1pts] README has setup/install instructions
- ✓ PASS [1pts] README documents environment variables
- ✓ PASS [1pts] README shows how to run agent
- ✓ PASS [1pts] README mentions prerequisites
- ✓ PASS [1pts] Demo video link in README

E3: Architecture diagram

- ✓ PASS [2pts] architecture_diagram exists
- ✓ PASS [1pts] agent_flowchart.png exists
- ✓ PASS [1pts] architecture.md documentation
- ✓ PASS [1pts] Docker/deployment config present

CATEGORY E SCORE: 15 / 15

CATEGORY F — Error Handling & Robustness [10 marks]

F1: Graceful API failure degradation

- ✓ PASS [1pts] Multiple try/except blocks (13 found)
- ✓ PASS [1pts] API error response function or message
- ✓ PASS [1pts] JIRAError handled in jira_tools.py

F2: Retry logic / LLM fallback

- ✓ PASS [1pts] OpenAI call wrapped in try/except
- ✓ PASS [1pts] ANSWER_NOT_FOUND fallback on LLM failure
- ✓ PASS [1pts] User-friendly fallback message

F3: No hard-coded secrets

- ✓ PASS [1pts] No API keys hardcoded in src/ files
- ✓ PASS [1pts] test_security.py exists

F4: Edge-case handling

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✓ PASS [1pts] Input validation in agent or workflow

✓ PASS [1pts] test_edge_cases.py exists

CATEGORY F SCORE: 10 / 10

CONFIDENCE CHECK — FINAL RESULTS

Cat A:  25/25 (100%)

Knowledge Base & RAG Pipeline

Cat B:  20/20 (100%)

Ticket Creation (Write Operation)

Cat C:  15/15 (100%)

Ticket Status Check (Read Operation)

Cat D:  15/15 (100%)

Agent Orchestration & Routing

Cat E:  15/15 (100%)

Code Quality & Documentation

Cat F:  10/10 (100%)

Error Handling & Robustness

TOTAL ESTIMATED SCORE: 100 / 100

🏆 EXCELLENT — Strong submission!

Checks passed : 80

Checks failed : 0

Items marked ✗ should be fixed before submission.